

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



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⚠ WARNING ⚠

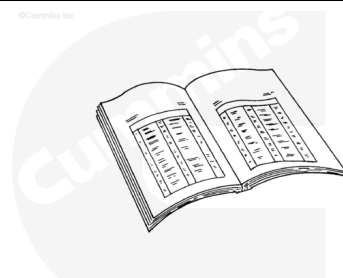
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the coolant thermostat housing. Refer to Procedure 008-013 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-013-tr.html>)
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)
- Remove the valve pushrods, injector pushrods, and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>)

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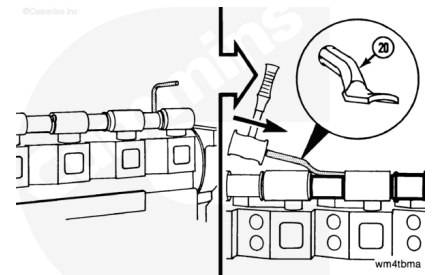
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- Remove the coolant thermostat housing. Refer to Procedure 008-013 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-013-tr.html>)
- Remove the injector supply lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)
- Remove and mark the valve pushrods so they can be installed in their original positions. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-001.html>)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-026.html>)

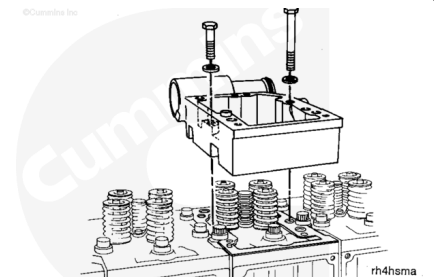
Remove

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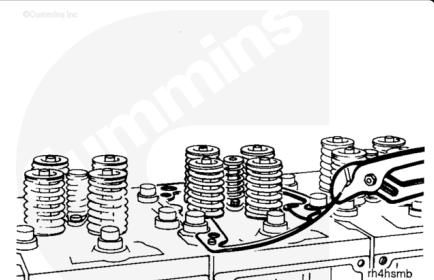
Use the water tube driver, Part Number ST-1319 (20), or equivalent, to drive the water transfer tube toward the front of the engine until the back part of the water transfer tube clears the rocker lever housing.



Remove the seven mounting capscrews and the rocker lever housing.



Remove and discard rocker lever housing gasket.



with Electronically Actuated Injector

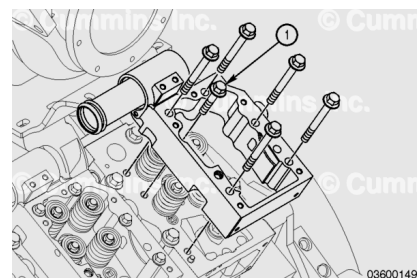
Use the water tube driver, Part Number ST-1319 (20), and a hammer to drive the water transfer tube toward the front of the engine until the back part of the water transfer tube clears the rocker lever housing.



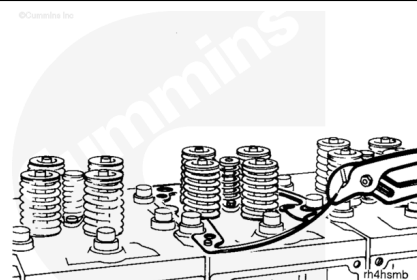
Remove the six mounting capscrews, plain washers, and the rocker lever housing.

Note : Note the placement of the single shorter capscrew (1).

All rocker lever housings are **not** the same. Tag or mark the parts with their respective locations relative to cylinder number to aid during assembly.



Remove and discard the gasket.



Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

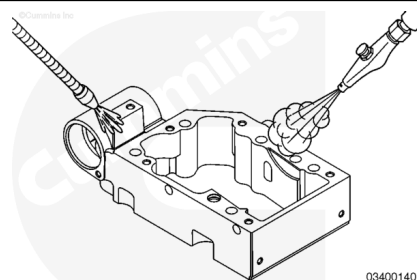
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

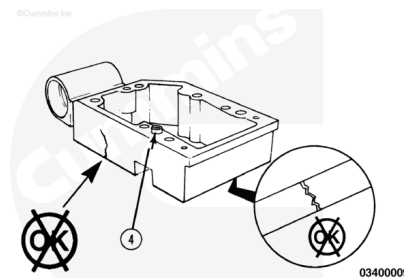


Use steam or solvent to clean the parts.

Dry with compressed air.

Inspect the parts for damage. If the ring dowel (4) is damaged, it **must** be replaced. Use a blind hole puller contained in light duty puller kit, Part Number 3375784, or equivalent, to remove the dowel ring.

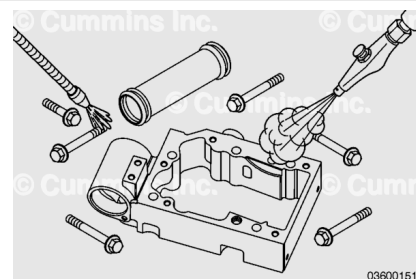
Use the crack detection kit, Part Number 3375432, to inspect the housing for cracks. If the housing is cracked, it **must** be replaced.



with Electronically Actuated Injector

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



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When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

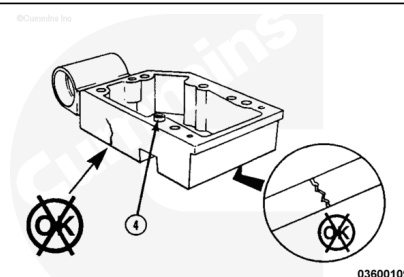
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam or solvent to clean the parts.

Dry with compressed air.

Use a nylon abrasive pad, Scotch-Brite™ 7448, or equivalent to thoroughly clean the rocker lever housing mating surface on the engine.

Inspect the parts for damage. If the ring dowel (4) is damaged, it **must** be replaced. Use a blind hole puller contained in light duty puller kit, Part Number 3375784; or equivalent, to remove the dowel ring.



Use the crack detection kit, Part Number 3375432, to inspect the housing for cracks. If the housing is cracked, it **must** be replaced.

Install

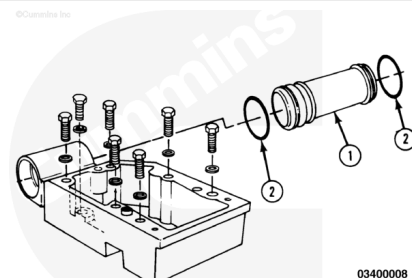
with Mechanically Actuated Injector

Lubricate the o-rings (2) with vegetable oil.

Note : Use of other lubricants, such as engine oil, will cause the o-rings to swell.

Install the o-rings into the grooves on the outside diameter at each end of the water transfer tubes. Do **not** overstretch, twist, or kink the o-rings.

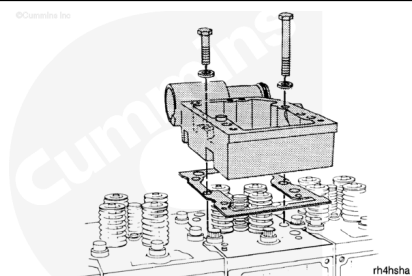
Install the water transfer tube (1).



Note : The gasket and housing must be aligned correctly with the dowel pins.

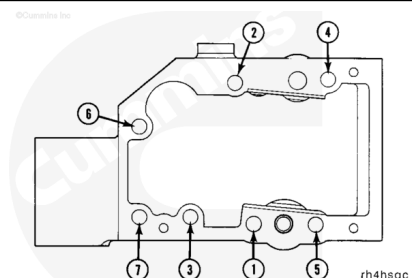
Install the gasket and the housing.

Install the seven capscrews. Hand-tighten.

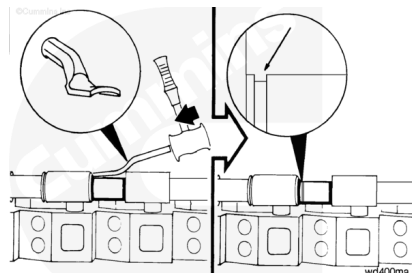


Use the sequence shown to tighten the capscrews.

Torque Value: 95 n•m [70 ft-lb]



Use water transfer tube driver, Part Number ST-1319, and a hammer to drive the water transfer tube into the adjacent housing until the water transfer tube is approximately centered.



with Electronically Actuated Injector

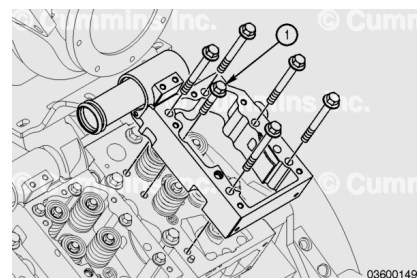
Lubricate the o-rings (2) with vegetable oil.

Note : Use of other lubricants, such as engine oil, will cause the o-rings to swell.

Install the o-rings into the grooves on the outside diameter at each end of the water transfer tubes. Do **not** overstretch, twist, or kink the o-rings.

Install the water transfer tube (1).

Make sure the tagged or labeled water transfer tubes are installed in the correct locations.

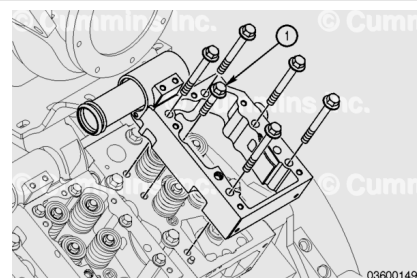


Install the new rocker housing gaskets to the cylinder heads while making sure the gaskets align correctly with both dowel pins and the rest of the bolt holes.

Make sure the tagged or labeled rocker lever housings are installed in the correct locations.

Check the threads of the rocker housing capscrews for damage. Check under the capscrew heads for cracks. Check for any deformation or necking of the capscrew.

Install the five long and one shorter (1) capscrew and washer. Hand-tighten them to a minimum of two full turns to obtain the correct thread engagement.

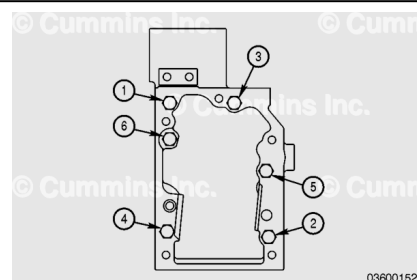


Use the sequence shown to tighten the capscrews.

Torque Value:

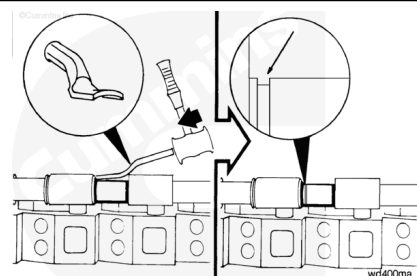
Rocker Lever Capscrews

1. 122 N•m [90 ft-lb]



⚠ CAUTION ⚠

Check that the o-rings are not cut, damaged or rolled over when driving the water transfer tubes into the adjacent rocker lever housings. Leakage will result if damaged.



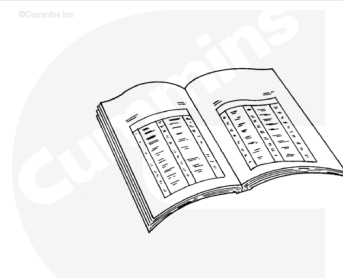
Use water transfer tube driver, Part Number ST-1319, and a hammer to drive the water transfer tube into the adjacent rocker lever housing until the water transfer tube is centered.

Finishing Steps

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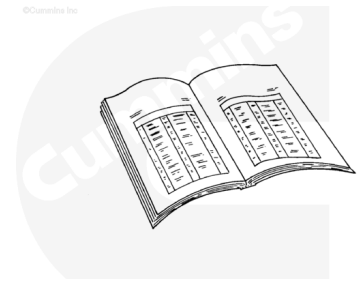
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- Install the valve pushrods, injector pushrods, and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>)
- Install rocker lever assembly. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html>)
- Adjust the overhead. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html>)
- Install rocker lever covers. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Install thermostat housing assembly. Refer to Procedure 008-013 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-013-tr.html>)
- Fill cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Operate the engine to 70°C [160°F] minimum coolant temperature and check for leaks.

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- Install the valve push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-004-014-tr.html>)
- Install the crossheads. Refer to Procedure 002-001 in Section 2. (</qs3/pubsys2/xml/en/procedures/20/20-002-001.html>)
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- Install rocker lever covers. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)
- Install the injector supply lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Install the coolant thermostat housing. Refer to Procedure 008-013 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-013-tr.html>)
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